

Acelero, Inc.

HELP AI Video Analysis Tool

Scope of Work — AI/ML Engineer

June 13 – July 31, 2026 · \$7,000

Project Context

Acelero, Inc. is developing an AI-assisted video analysis tool to support early childhood developmental assessment using the HELP 0–3 framework. A working prototype was built in 2025 as part of an IRB-approved study. This engagement takes that prototype further — making it more automated, more usable, and ready to support educator workflows at scale.

This project has two contractor roles. An Applied AI Scientist is leading the AI approach — exploring how AI can most usefully support HELP assessment practice, improving prompt performance, and defining what success looks like. The AI/ML Engineer builds the technical infrastructure that brings that approach to life. The two roles work closely together throughout.

A note on the AI approach: the existing prototype attempted to have AI independently score videos and measure agreement with human raters. Based on deeper analysis, the approach is being refined. Rather than replacing human judgment, the AI will surface potentially overlooked skill credits for educator review — a more tractable and defensible design. The Applied AI Scientist is leading this redesign. The engineer's job is to build a system that makes this new approach run reliably at scale.

Your Role

Build and implement the technical components needed to support the AI-assisted assessment workflow. This includes the processing pipeline, review interface, data infrastructure, and supporting tools needed to test and use the approach efficiently.

You will collaborate closely with the Applied AI Scientist and Acelero's content lead throughout. The Scientist determines the AI approach and what to build. You build it and make it run. Weekly three-way check-ins are expected for the duration of the engagement.

Core Scope — Pipeline & Review Workflow

Target completion: July 31, 2026

Example Activities

- Build a workflow for processing batches of observation videos
- Connect video inputs, child demographic information, and Gemini API processing
- Generate structured outputs that organize AI observations and credit suggestions by skill ID, domain, strand, confidence score, and timestamped evidence
- Build a simple review interface where educators can view AI-surfaced suggestions and accept or reject each one
- Capture reviewer decisions in a structured, exportable format for analysis and future model improvement
- Support storage and export of assessment outputs in standard formats (CSV at minimum)
- Participate in regular project check-ins with the Scientist and Acelero team

Deliverables

- Functional batch-processing pipeline
- Structured output format for AI-generated observations and credit suggestions
- Human review interface with accept/reject functionality and low-confidence flagging
- Feedback logging and export capability
- Documentation sufficient to support handoff and future development

Potential Other Activities

These activities depend on both pace and external factors including HELP Connect platform readiness.

- Exploring cloud-based deployment options (AWS or equivalent)
- Preparing API endpoints or integration pathways for future connection to external platforms
- Laying groundwork for future HELP Connect integration
- Deployment and operations documentation

Indicators of Progress

- Videos can be processed in batches without manual intervention per video
- Outputs are structured, retrievable, and consistent across runs
- Educators can review AI suggestions and log their decisions through the interface
- Reviewer feedback is captured and exportable in a usable format
- Documentation is sufficient for Acelero's team to operate and build on the system

What Acelero Will Provide

- The existing Gemini-powered prototype and full prompt
- IRR study dataset and reliability results
- Child demographic records and video files

- Direction from the Applied AI Scientist on workflow design and output specifications
- Access to relevant source materials, documentation, and GitHub repository
- Weekly three-way check-ins throughout the engagement

Payment Schedule

Payment is milestone-based, tied to deliverable completion and approval.

Milestone	Target Date	Payment
Project kickoff and planning	June 13	\$500
Initial processing pipeline operational	June 30	\$1,500
Integration with AI workflow complete	July 15	\$1,500
Review interface and feedback logging complete	July 20	\$1,500
Final delivery and documentation	July 31	\$2,000
Total		\$7,000

What We Are Looking For

If you have not already submitted a proposal, please include:

- Your proposed technical approach for the pipeline and review interface
- Recommended deployment path with rationale
- Relevant experience with production LLM pipelines, Gemini or equivalent multimodal APIs, and structured output generation
- At least one example from a comparable project, with a reference
- Confirmation that you can meet the July 31 deadline